Path Codes on Shared Knowledge Graphs

Part 0: Model, Definitions and Running Example

R. Sandoval Aréchiga (working notes)

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Abstract

These notes fix the model and notation used by all subsequent results. A transmitter and a receiver share a labelled directed graph; a meaning is conveyed by a walk on it. The walk-to-meaning map is a synonymous mapping in the sense of Niu and Zhang [1], but here it is *induced by the graph structure*. We define path codes with and without decoder memory, the misalignment between transmitter and receiver graphs, and the geodesic semantic distortion. A ten-node example is worked out in full.

1 Standing assumptions

Assumption 1 (Finite rooted DAG)**.** *The graph is finite, directed and acyclic, with a unique root r and a non-empty set of terminal nodes (nodes with out-degree 0). Every node is reachable from r.*

Assumption 2 (Node-Markov routing)**.** *At every non-terminal node v the next edge is drawn from a distribution $p(\cdot \mid v)$ that depends on v only.*

Assumption 3 (Meaning at completion)**.** *A meaning is attached to complete walks (from r to a terminal), not to prefixes.*

Assumption 1 guarantees that the set of complete walks is finite, so all quantities below are finite sums. Knowledge graphs usually contain cycles; if walks are bounded in length by L, unrolling the graph into L layers yields a DAG that satisfies Assumption 1, and all results apply to it. Assumption 3 is relaxed in the hierarchical setting of Part 3, where prefixes carry coarse meanings.

2 Definitions

Definition 1 (Trajectory network)**.** *A trajectory network is a tuple $G = (V, E, r, T, \ell, p)$ where (V, E) satisfies Assumption 1, $T \subset V$ is the set of terminals, $\ell : V \to \Sigma$ is a labelling with values in a finite alphabet Σ, and p is a routing kernel: for every $v \in V \setminus T$, $p(\cdot \mid v)$ is a probability distribution on the out-neighbours of v.*

Definition 2 (Walks and path measure)**.** *A* complete walk *is a sequence $\pi = (v_0, v_1, \ldots, v_k)$ with $v_0 = r$, $(v_{t-1}, v_t) \in E$ and $v_k \in T$. Let Π_G denote the set of complete walks. The routing kernel induces the* path measure

$$P(\pi) = \prod_{t=1}^{k} p(v_t \mid v_{t-1}), \qquad \pi \in \Pi_G. \tag{1}$$

A random complete walk with law P is denoted Π.

Definition 3 (Meaning map, preimage, synonymy)**.** *A* meaning map *is a surjection $\mu : \Pi_G \to \mathcal{M}$ onto a finite set of meanings. The* preimage *of $m \in \mathcal{M}$ is $\Pi(m) = \mu^{-1}(m)$. The random meaning is $M = \mu(\Pi)$, with law*

$$P_M(m) = \sum_{\pi \in \Pi(m)} P(\pi). \tag{2}$$

Two walks are synonymous *if they have the same image under μ. The network exhibits synonymy if $|\Pi(m)| \geq 2$ for some m.*

Remark 1 (Relation to synonymous mappings). The preimages $\{\Pi(m)\}_{m \in \mathcal{M}}$ partition Π_G, so μ is a synonymous mapping in the sense of [1, Def. 2.1], with walks playing the role of syntactic symbols. In particular, $H(M)$ is their semantic entropy H_s, and coding each meaning directly achieves it [1, Thm. 6.8], [2]. Two features have no counterpart in the unstructured setting: (i) a walk is produced and consumed *decision by decision*, so a decoder may or may not keep memory of the prefix; (ii) transmitter and receiver may hold *different* networks.

Because μ is a function of Π, the chain rule gives the decomposition used throughout:

$$H(\Pi) = H(M) + H(\Pi \mid M). \tag{3}$$

Definition 4 (Path codes)**.** *A* path code *for (G, μ) is a pair $(f, \mathcal{C})$ where $f : \mathcal{M} \to \Pi_G$ is a selection with $f(m) \in \Pi(m)$, and $\mathcal{C}$ is a sequential binary code that describes $f(m)$ one decision at a time, so that the receiver reconstructs the walk while traversing it. The code is*

- native *if the coding distribution used for the decision at v_{t-1} is the shared kernel $p(\cdot \mid v_{t-1})$;*
- adapted *if it uses a kernel agreed for the code, which may differ from p; an adapted code is* memoryless *if that kernel depends on the current node only, and* with memory *if it may depend on the prefix $(v_0, \ldots, v_{t-1})$.*

The rate of a path code is its expected length $L = \mathbb{E}[\,|\mathcal{C}(f(M))|\,]$.

Lemma 1 (Cost of a walk; arithmetic coding [3, 4, 5])**.** *For any distribution Q on Π_G that factorises as a product of sequential conditionals, there is a prefix-free sequential code with $|\mathcal{C}(\pi)| < -\log_2 Q(\pi) + 2$ for every π with $Q(\pi) > 0$. In particular, a native path code spends $-\log_2 p(v_t \mid v_{t-1})$ bits per decision, up to 2 bits overall, and a node with out-degree 1 costs 0 bits.*

Definition 5 (Transmitter and receiver networks; misalignment)**.** *The transmitter holds G_S with routing p_S and the receiver holds G_R with routing q_R, over a common node set V and a common meaning map μ. Let P_S and Q_R be the induced path measures. The* misalignment *is*

$$A = D(P_S \,\|\, Q_R) = \sum_{\pi} P_S(\pi) \log_2 \frac{P_S(\pi)}{Q_R(\pi)} \in [0, \infty]. \tag{4}$$

An edge missing in G_R is encoded as $q_R(e \mid v) = 0$. The edited nodes *are $V_\Delta = \{v : p_S(\cdot \mid v) \neq q_R(\cdot \mid v)\}$.*

Definition 6 (Visit measure)**.** *$\nu_S(v) = \Pr_{P_S}[\, v \in \Pi\,]$ is the probability that a walk drawn from P_S visits v. In a DAG each node is visited at most once, so $\nu_S(v)$ also equals the expected number of visits.*

Definition 7 (Geodesic semantic distortion)**.** *Let O be a connected undirected graph on $\mathcal{M}$ (an ontology or concept graph). The* geodesic distortion *is $d_O(m, \hat{m})$, the length of a shortest path between m and $\hat{m}$ in O. It is* ultrametric *if $d_O(m, m'') \leq \max\{d_O(m, m'), d_O(m', m'')\}$ for all triples, as happens on the leaves of a rooted tree of height h with $d(m, m') = h - \mathrm{depth}(\mathrm{lca}(m, m'))$.*

Definition 8 (Quotients)**.** *A partition* $\mathcal{Q} = \{B_1, \dots, B_J\}$ *of* $\mathcal{M}$ *is a* quotient codebook*. Its diameter is* $\operatorname{diam}(\mathcal{Q}) = \max_j \max_{m,m' \in B_j} d_O(m, m')$*. Given representatives* $b_j \in B_j$*, the decoder outputs* b_j *whenever* $M \in B_j$*. The quotient rate is* $H(\mathcal{Q}) = -\sum_j P_M(B_j) \log_2 P_M(B_j)$*.*

3 Statements proved in Parts 1–4 (for orientation only)

- **Part 1 (price of memory).** Native path codes have rate $H(M) + \mathbb{E}[g(M)]$, where $g(m) = \log_2\big(P_M(m) / \max_{\pi \in \Pi(m)} P(\pi)\big)$ and $0 \le \mathbb{E}[g(M)] \le H(\Pi \mid M)$. Adapted codes attain $H(M)$; they need no decoder memory when the meaning is endpoint-determined, and may need it when the meaning depends on the route. Bits-back [6] attains $H(M)$ in an amortised sense.
- **Part 2 (structural misalignment).** $A = \sum_v \nu_S(v)\, D\big(p_S(\cdot \mid v) \,\|\, q_R(\cdot \mid v)\big)$, with two-sided local bounds; A splits into a meaning-level term, the semantic relative entropy of [1, Def. 4.1], and a phrasing-level term. Routing, topological and label edits respectively cost bits, are detected, and cause silent errors.
- **Part 3 (geodesic rate–distortion).** Quotient codes achieve points of the rate–distortion region, are tight at both ends and strictly suboptimal in between. Successive refinability under ultrametric distortion holds numerically for uniform sources (conjecture) and fails for non-uniform ones.
- **Part 4 (Weaver's levels).** Technical, semantic and effectiveness levels form a chain of quotients with rates $H(\Pi) \ge H(M) \ge H(a(M))$.

4 Running example: "what day is today"

Network. Ten nodes (Fig. 1). From the root r a single edge leads to c, "according to the calendar". At c the speaker chooses a frame: a = "today is" with probability 0.6, or b = "tomorrow is" with probability 0.4. From a the walk ends at Monday, Tuesday or Wednesday with probabilities $(0.5, 0.3, 0.2)$. From b it ends at Tuesday′, Wednesday′ or Thursday with the same probabilities. The meaning μ is *the day it is today*, so "today is Monday" and "tomorrow is Tuesday" are synonymous. The synonymy is induced by the graph: it comes from the structure of the two frames, not from a list.

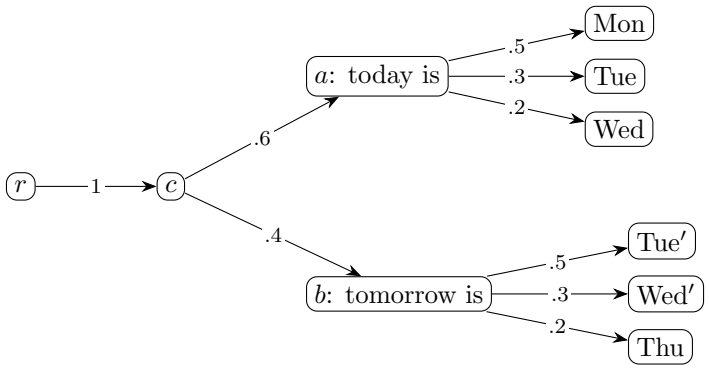


Figure 1: Running example. Meaning = the day it is today; "today is Mon" and "tomorrow is Tue′" are synonymous.

Path and meaning laws.

Walk π	$P(\pi)$	$\mu(\pi)$	$-\log_2 P(\pi)$
$r\,c\,a\,$Mon	0.30	Monday	1.737
$r\,c\,a\,$Tue	0.18	Tuesday	2.474
$r\,c\,a\,$Wed	0.12	Wednesday	3.059
$r\,c\,b\,$Tue$'$	0.20	Monday	2.322
$r\,c\,b\,$Wed$'$	0.12	Tuesday	3.059
$r\,c\,b\,$Thu	0.08	Wednesday	3.644

Hence $P_M = (0.5, 0.3, 0.2)$ for (Monday, Tuesday, Wednesday), $H(\Pi) = 2.456$, $H(M) = 1.485$ and $H(\Pi \mid M) = 0.971$ bits, in agreement with (3).

Cost of a walk (Lemma 1). The walk $r\,c\,a\,$Mon costs $0 + 0.737 + 1 = 1.737$ bits. The first decision is free because r has out-degree 1.

Price of memory. For every meaning, the most probable walk goes through a, and

$$\frac{P_M(m)}{\max_{\pi \in \Pi(m)} P(\pi)} = \frac{1}{0.6} \quad \text{for all } m.$$

Thus $g(m) = \log_2(1/0.6) = 0.737$ bits for all m, and the best native code has rate $L_0 = 2.222 = H(M) + 0.737$ bits. The excess is exactly the cost of the decision at c between "today is" and "tomorrow is", which carries no meaning. An adapted code knows that the selected walks always pass through a, codes that decision with probability 1, and attains $H(M) = 1.485$; here it needs no memory (Part 1). Here $\mathbb{E}[g(M)] = 0.737 < H(\Pi \mid M) = 0.971$: the upper bound of Part 1 is not tight because the conditional law of walks given the meaning is $(0.6, 0.4)$, not uniform.

Misalignment (Part 2).

- *Probability edit.* Suppose the receiver routes uniformly at a, $q_R(\cdot \mid a) = (1/3, 1/3, 1/3)$, and agrees elsewhere. Then $V_\Delta = \{a\}$, $\nu_S(a) = 0.6$, and

  $$A = 0.6 \cdot D\big((0.5, 0.3, 0.2) \,\|\, (\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3})\big) = 0.060 \text{ bits},$$

  below the bound $0.6 \log_2 3 = 0.951$. It splits as 0.036 bits at the level of meanings plus 0.024 bits that only path codes pay.

- *Missing edge.* If the receiver lacks the edge $b \to$ Thu, the walk $r\,c\,b\,$Thu has $Q_R = 0$ and $A = \infty$ under the full path measure: the receiver detects an impossible walk with no checksum. An adapted code that always selects walks through a never uses that edge. The structural mismatch is invisible to it.

Geodesic distortion (Part 3). Let O be the path Monday–Tuesday–Wednesday, so $d_O(\text{Mon}, \text{Tue}) = d_O(\text{Tue}, \text{Wed}) = 1$ and $d_O(\text{Mon}, \text{Wed}) = 2$. The quotient $\{\{\text{Mon}, \text{Tue}\}, \{\text{Wed}\}\}$ has diameter 1 and rate $H(0.8, 0.2) = 0.722$ bits. With the best representative (Monday) its expected distortion is 0.3. The alternative $\{\{\text{Mon}\}, \{\text{Tue}, \text{Wed}\}\}$ also has diameter 1, with rate 1.000 bit and distortion 0.2. Part 3 compares both with the Blahut–Arimoto value of $R(D)$ [7]: at $D = 0.3$, $R(D) = 0.439$ bits.

References

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